

Professional Summary :

- Over **7 years** of experience in analyzing, designing, developing, and implementing data architectures and frameworks as a Data Engineer across various cloud platforms (AWS, GCP, Azure).
- Expertise in Data Warehousing and Decision Support Systems, with extensive experience in implementing full lifecycle Data Warehousing projects.
- Proficient in Hadoop Architecture and its ecosystem components, including HDFS, MapReduce, Hive, HBase, Oozie, Sqoop, and Zookeeper.
- Strong experience in big data storage, querying, processing, and analysis, with hands-on skills in Hadoop, Spark (Scala/Python), and Apache Beam.
- Hands-on experience in building ETL pipelines and visualizations on cloud platforms such as AWS, Azure, GCP, and Databricks, using tools like Azure Data Factory, Dataflow, and BigQuery.
- Designed and implemented cloud migration strategies on Azure and GCP, leveraging services like Azure SQL Database, Azure Data Lake, Cloud Storage, and BigQuery.
- Skilled in developing and deploying Docker and Kubernetes-based applications across multiple cloud environments, enabling seamless CI/CD workflows.
- Proficient in relational databases (Oracle, SQL Server, MySQL) and NoSQL databases (MongoDB, Cassandra, HBase), as well as working with data visualization tools like Power BI and Tableau.
- Strong programming experience in Python, Scala, and Kafka, with a focus on machine learning models such as Linear and Logistic Regression.
- Experienced with automation tools (Ansible, Jenkins, Chef) and schedulers like Oozie, Airflow, and Crontab for managing data pipelines and deployments.

Technical Skills:

Big Data Technologies	HDFS, YARN, Map Reduce, Hive, Pig, Impala, Sqoop, Storm, Flume, Spark, Apache Kafka, Zookeeper, Ambari, Oozie, MongoDB, Cassandra, Mahout, Puppet, Avro, Parquet, Snappy, Falcon.
NO SQL Databases	Postgres, HBase, Cassandra, MongoDB, Amazon Dynamo DB, Redis
Hadoop Distributions	Cloudera (CDH3, CDH4, and CDH5), Hortonworks, MapR, and Apache.
Languages	Scala, Python, R, XML, XHTML, HTML, AJAX, CSS, SQL, PL/SQL, HiveQL, Unix, Shell Scripting
Source Code Control	GitHub, CVS, SVN,
Cloud Computing Tools	Amazon AWS, Microsoft Azure, GCP
Databases	Teradata Snowflake, Microsoft SQL Server, MySQL, DB2
DB languages	MySQL, PL/SQL, PostgreSQL, R-SQL, T-SQL & Oracle
Build Tools	Jenkins, Maven, Ant, Log4j
Business Intelligence Tools	Tableau, Power BI
Development Tools	Eclipse, IntelliJ, Microsoft SQL Studio, NetBeans
ETL Tools	Talend, Pentaho, Informatica, Ab Initio, SSIS
Development Methodologies	Agile, Scrum, Waterfall, V model, Spiral, UML

Professional Experience:

Senior Data Engineer - CA.

Client: Nerd wallet | Domain: Financial Services

Jan 2023 –Present

Responsibilities:

- Developed multiple data pipelines, end-to-end ETL, and ELT processes for data ingestion and transformation in Google Cloud Platform (GCP), coordinating team tasks.
- Transferred data from on-premises Cloudera to GCP, working with file formats like Parquet, Avro, and CSV.

- Designed and implemented various layers of Data Lake and created star schemas in BigQuery.
- Created and managed GCP Data Proc clusters for data processing and configuration.
- Built real-time data streaming pipelines using GCP Pub/Sub and Dataflow for data ingestion and processing.
- Implemented data transformation and enrichment processes within Pub/Sub pipelines using Apache Beam and Spark Streaming for real-time data processing.
- Applied encryption, access controls, and data masking techniques to secure sensitive data in Pub/Sub pipelines.
- Designed, developed, and maintained scalable data pipelines on GCP using Java and Spark.
- Developed Spark programs for data transformation, loading refined data into partitioned tables in enterprise data warehouses.
- Utilized Pentaho Reporting for generating interactive reports and dashboards from multiple data sources.
- Ingested data from various sources into cloud storage systems and managed cloud security and data encryption.
- Built machine learning models using Python and created visualizations with Dash by Plotly.
- Provisioned virtual machines in AWS and Azure, deploying applications across environments with Ansible automation.
- Developed batch and streaming jobs using GCP services like BigQuery, Pub/Sub, Dataflow, DataProc, Cloud Run, Compute Engine, and Cloud Composer.
- Built a reporting data warehouse from ERP systems, focusing on Order Management, Invoice, and Service Contracts modules.
- Created real-time streaming applications using PySpark, integrated with Kafka, and persisted data in NoSQL databases like HBase and Cassandra.
- Configured GCP services like DataProc, Dataflow, Pub/Sub, BigQuery, and Cloud Storage using Cloud SDK for efficient cloud operations.
- Developed data pipelines in GCP Cloud Composer using Airflow to manage ETL-related workflows.
- Implemented batch and streaming data ingestion pipelines in PySpark, processing data from various sources into HDFS Data Lake.
- Worked extensively with Big Data Hadoop clusters, integrating them with technologies like Spark, Sqoop, and Hive for large-scale data processing.

Data Engineer, Plano, TX

Client: Corewell Health | Domain: Healthcare

Aug 2021 – Dec 2022

Responsibilities:

- Optimized performance and enhanced algorithms in Hadoop using Spark Context, Spark SQL, Data Frame, and Spark Yarn.
- Worked extensively with Azure BLOB and Azure Data Lake Storage, loading data into Azure SQL Synapse Analytics.
- Managed Azure Data Lakes (ADLS) and Data Lake Analytics, integrating them with various Azure services for seamless operations.
- Automated Azure Databricks activities, including running notebook activities and scheduling jobs via Azure Data Factory Pipelines.
- Leveraged Databricks for cluster management, job monitoring, and implemented external Delta Tables in Databricks and Snowflake.
- Utilized Azure Cosmos DB for globally distributed data across multiple Azure regions, ensuring high availability and throughput.
- Developed data partitioning strategies in Databricks for daily ingestion tasks and automated data handling between Raw and Curated Layers.
- Designed end-to-end data ingestion pipelines using Azure Databricks and integrated Azure services such as Azure Data Factory, Spark SQL, and U-SQL.
- Developed and supported Ab Initio packages for Teradata encryption applications in large-scale mortgage data marts.
- Automated Databricks cluster scaling and spark job submissions for efficient resource utilization.
- Worked with Azure Purview for advanced data governance, implementing features like Data Map, Data Catalog, and Data Insights for secure data management.
- Integrated Azure Automation for managing processes across various Azure services such as virtual machines (VMs), SQL databases, and Azure Active Directory.
- Architected real-time analytics platforms on Azure using Spark Structured Streaming and Databricks for massive data processing.
- Migrated on-premises data from SQL Server to Azure Synapse Analytics, working on data transformation and integration with Azure Data Lake and Storage.
- Created IoT solutions on Azure Stack Hub, ensuring reliable communication between IoT devices and on-premises systems.
- Involved in designing and managing DataStage ETL jobs, collaborating with developers to ensure data consistency throughout the ETL processes.
- Used Apache Kafka to upload clickstream data to HDFS and leveraged Spark Streaming API for real-time micro-batch processing.
- Extensively used AWS S3, Glue, and Redshift for cloud-based data processing, archiving, and retention management.
- Integrated traditional ETL processes with Hadoop, accelerating data extraction, transformation, and loading for structured and unstructured data.

- Managed containerized applications using Docker and Kubernetes, automating builds on Docker Hub and integrating with AWS cloud services like EMR, Athena, and Snowflake.

Role: Azure Data Engineer, India

Client: AXA

Oct 2019 – Jul 2021

Responsibilities:

- Involved in the full project lifecycle, including data analysis, preprocessing, user-generated data extraction, and modeling for various big data projects.
- Developed data pipelines in Airflow on GCP for ETL jobs, leveraging Spark applications written in both Scala and Python.
- Implemented data quality checks and validation rules using Scala to ensure data accuracy, completeness, and consistency.
- Experienced with GCP services like BigQuery, Cloud Functions, and Data Proc, using tools like Kafka, Spark, HBase, Hive, and HDFS for analyzing massive datasets.
- Hands-on experience using Cloudera for managing Hadoop environments and working with various file formats (CSV, Parquet, ORC) for data processing.
- Created Spark applications to load large volumes of data into Hive ORC tables and built Snowpipe for continuous data loading using NiFi in Snowflake.
- Proficient in using Snowflake for data ingestion, collaboration, and leveraging MPP clusters for data processing tasks like loading, transforming, and querying.
- Designed and implemented ETL processes using Scala and Apache Spark for cleaning, transforming, and aggregating raw data for analysis.
- Built POCs for migrating projects from on-premises Hadoop MapR systems to GCP and developed SSIS packages for data transfers.
- Configured GCP services like Data Proc, Storage, and BigQuery, using the Cloud Shell SDK and worked on PySpark implementations for complex data transformations.
- Implemented Kafka models to extract recent data into external Hive tables and worked with Java scripts for data formatting in Hadoop environments.
- Participated in data acquisition projects using tools like Sqoop, Pig, Flume, and HDFS for extracting historical and real-time data.
- Expertise in creating custom Spark SQL aggregate functions and UDFs in Hive, used for interactive querying and complex data manipulations.
- Applied machine learning techniques, including Linear and Logistic Regression, Decision Trees, and clustering algorithms (K-means, Hierarchical) using Python libraries like Pandas, NumPy, Scikit-learn, and SciPy.
- Created interactive data visualizations using Tableau and Power BI, optimized SQL queries, and collaborated with Business Analysts to design dashboards and generate reports.

Mindtree, INDIA

March 2018 to Sep 2019

Role: Data Engineer/ Data analyst

Responsibilities:

- Collaborated with business user's/product owners/developers to contribute to the analysis of functional requirements.
- Implemented Spark SQL queries that combine hive queries with Python programmatic data manipulations supported by RDDs and data frames.
- Used Kafka Streams to Configure Spark streaming to get information and then store it in HDFS.
- Extract Real-time feed using Spark Streaming and convert it to RDD and process data in the form of Data Frame and save the data in HDFS.
- Developing Spark scripts, UDFS using Spark SQL query for data aggregation, querying, and writing data back into RDBMS through Sqoop.
- Installed and configured Hadoop MapReduce HDFS Developed multiple MapReduce jobs in java for data cleaning and preprocessing.
- Installed and configured Pig and also written Pig Latin scripts. Write MapReduce job using Pig Latin.
- Worked on analyzing Hadoop clusters using different big data analytic tools including HBase database and Sqoop.
- Worked on importing and exporting data from Oracle, and DB2 into HDFS and HIVE using Sqoop for analysis, visualization, and generating reports.
- Creating and inserting data into Hive tables for dynamically inserting data into data tables using partitioning and bucketing for EDW tables and historical metrics.
- Experienced in handling large datasets using Partitions, Spark in Memory capabilities, Broadcasts in Spark, Effective & efficient Joins, Transformations, and others during the ingestion process itself.
- Created ETL packages with different data sources (SQL Server, Oracle, Flat files, Excel, DB2, and Teradata) and loaded the data into target tables by performing different kinds of transformations using SSIS.
- Designed and developed data integration programs in a Hadoop environment with No SQL data store Cassandra for data access and analysis.

- Created partitions, bucketing across the state in Hive to handle structured data using Elastic search.
- Performed Sqoop for various file transfers through the HBase tables for processing of data to several No SQL DBs- Cassandra, Mongo DB.

Educational Qualifications:

Bachelor's of Technology in Computer Science in JECRC University, Jaipur
Masters of Science in Computer Science from California State University

2014 to 2018
2021 to 2023